Bulk cloud model

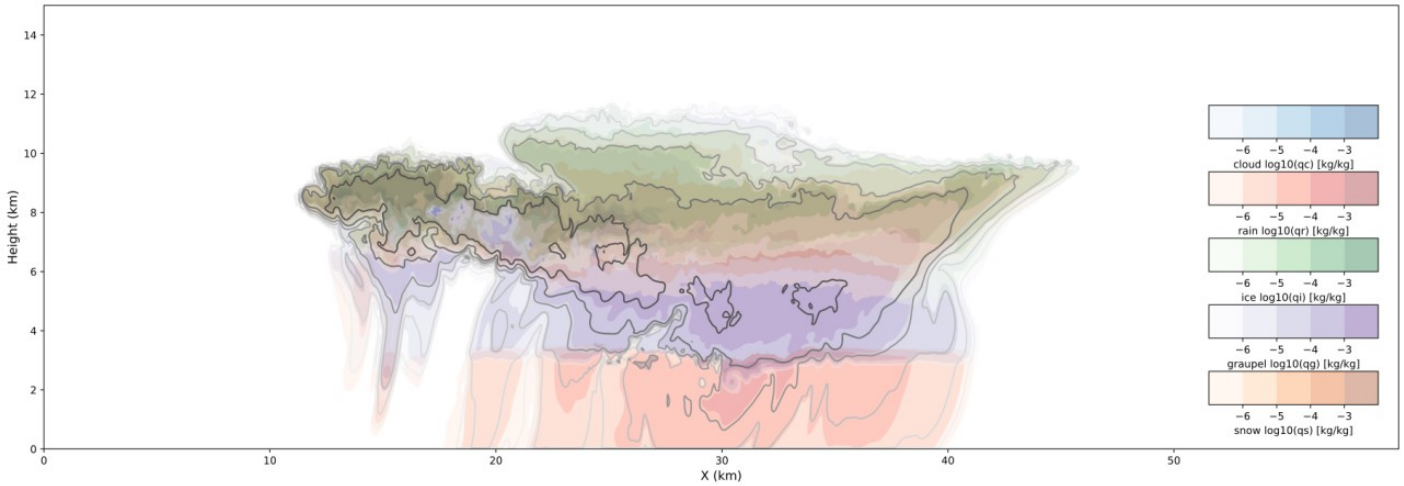
palm group

Institute of Meteorology and Climatology, Leibniz Universität Hannover

Bulk cloud model

Overview

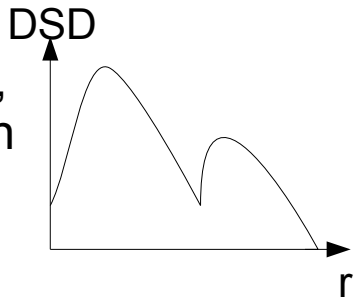
- The Bulk cloud model (BCM) is able to simulate warm and mixed-phase clouds

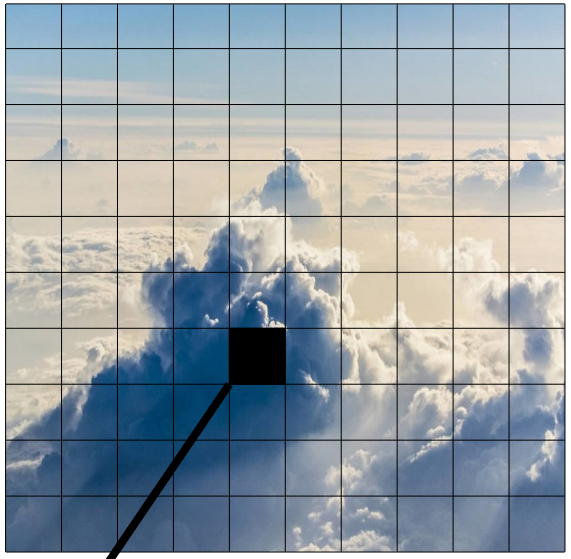


Bulk cloud model

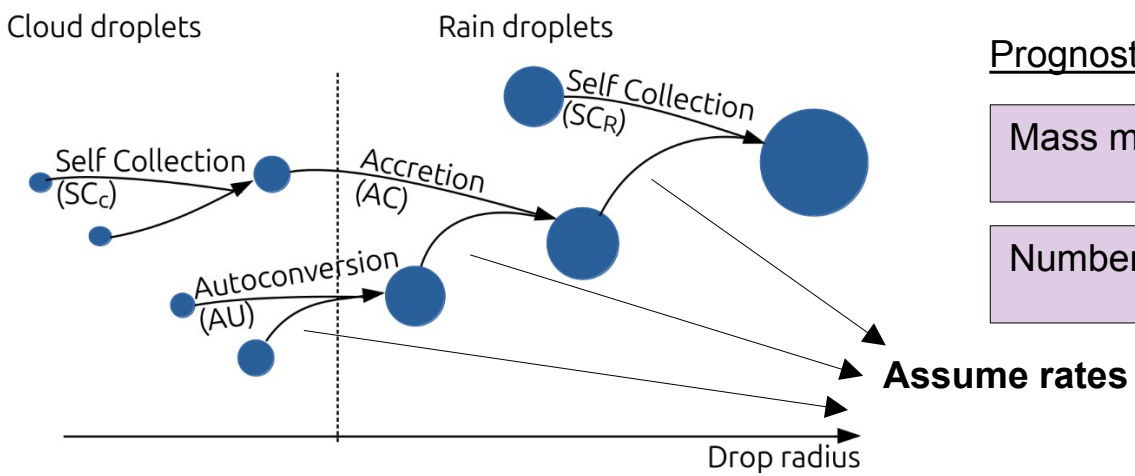
Physics

- Assumption that cloud droplets, rain droplets or ice particles can be described by a **given distribution function**
- Different number of **moments** can be considered for different number of **species**
- Microphysical processes are **parameterized**





For each grid box



Prognostic quantities

Mass mixing ratios

Number concentrations

Number of moments

Bulk cloud model

Physics II

- Available schemes:
  - saturation adjustment
  - Kessler (1969)
  - Seifert and Beheng (2001,2006)
  - Morrison et al. (2007)

- Implementation of liquid phase based on the code of *DALES & UCLA-LES* and ice phase on *ICON*

- Usage of mixing ratios

- Underlying prognostic equations of liquid water potential temperature (`pt`) and total water content (q) (e.g., *Emanuel, 1994*)

$$\theta_L \approx \theta - \left(\frac{\theta}{T}\right)\left(\frac{L_v}{c_{pd}}\right)(r_l),$$

Bulk cloud model

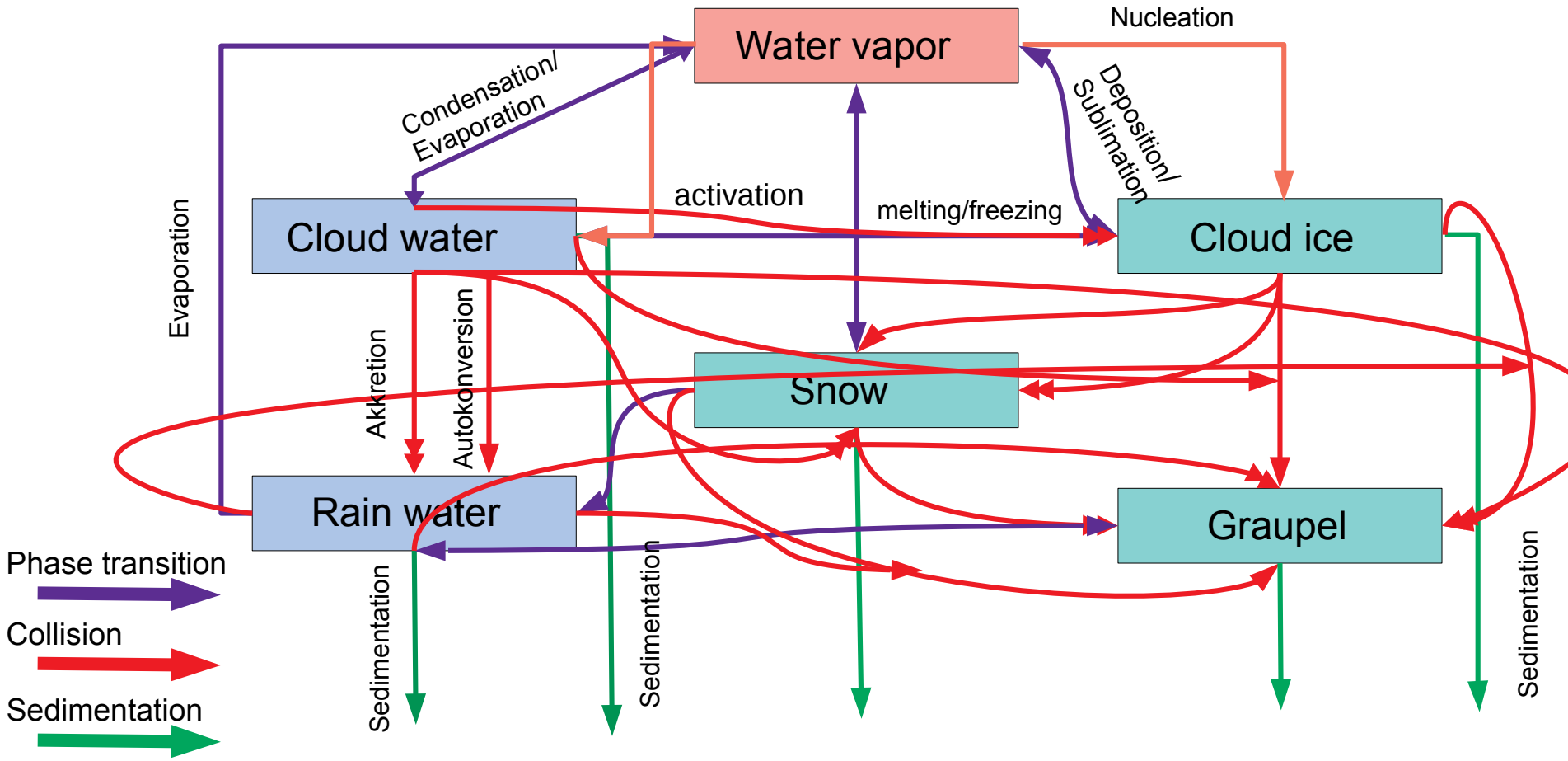
Not included Physics

Examples of cloud processes not taken into account:

- Effects of mixing / entrainment on the cloud droplet distribution
- Effects of turbulence on coalescence
- Different coalescence efficiencies
- Collisional breakup
- Ice nucleation only roughly parametrized

Bulk cloud model

Physics parameterized



&bulk_cloud_parameters

General parameters

`cloud_scheme`	choose cloud scheme
`nc_const`	number of cloud droplets [m^{-3}]
`na_init`	number of aerosol [m^{-3}]
`cloud_water_sedimentation`	flag for sedimentation of clod droplets
`microphysics_ice_phase`	flag to turn on ice phase
`snow/graupel`	flag to additionally turn on snow and graupel species (only allowed together)

Find all parameters at latest changes:

https://docs.palm-model.org/23.04/Reference/LES_Model/Namelists/#bulk-cloud-parameters

https://palm.muk.uni-hannover.de/trac/wiki/doc/app/bcmequ

Bulk cloud model

Output quantities

Output for cloud quantities is steered via:

`&runtime_parameters`

`Timeseries (default)`

`Lwp`	liquid water path
`cwp/rwp/iwp/gwp/swp`	cloud/rain/ice/graupel/snow water path

`2D/3D output`

`ql/qc/qr/qg/qi/qs`	Mixing ratios of: liquid water, cloud water, rain water, graupel, ice, and snow
`nc/nr/ng/ni/ns`	Number concentration of: cloud droplets, rain droplets, graupel, ice, and snow
`prr`	Total precipitation rate
`prr_cloud, prr_rain, prr_ice, prr_graupel, prr_snow`	Precipitation rates of: cloud, rain, ice, graupel, and snow

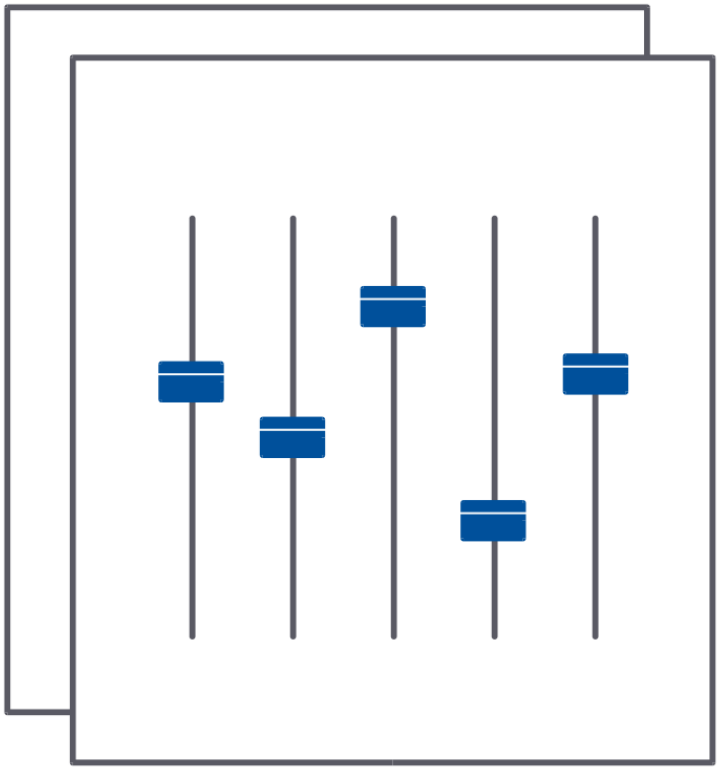
`Profiles`

- `Same quantities as for 2D/3D output`

Bulk cloud model

What must be considered for simulations with clouds?

- Add Namelist `&bulk_cloud_parameters` in Job-file.
- Set temperature and humidity profile in a way that air becomes supersaturated at some point
- Set a `cloud_scheme`
- Add output of cloud quantities
- Do you want to have radiation effects included? Yes: Turn on RRTMG



Bulk cloud model

Example cases

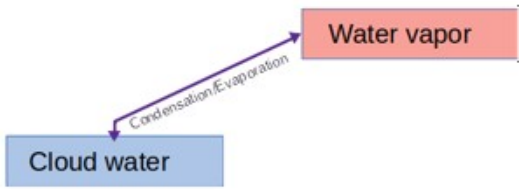
- Examples can be found in the test repository
- `../model/tests/cases/`
- `mixed-phase-stratus_with_rrtmg`

```
!-----------------------------------------------------------------------------
!-- BULK CLOUD PARAMETER NAMELIST
!   Documentation: https://palm.muk.uni-hannover.de/trac/wiki/doc/app/bcmpar
!-----------------------------------------------------------------------------
&bulk_cloud_parameters
!
!-- set microphyiscs scheme
!-----------------------------------------------------------------------------
   cloud_scheme            = 'seifert_beheng',        ! two-moment liquid water microphysics
                                                        ! including autoconversion,
                                                        ! accretion, sedimentation, precipitation
!
!-- steering of liquid phase
!-----------------------------------------------------------------------------
   collision_turbulence              = .TRUE.,              ! parametize turbulence effects for collision
   cloud_water_sedimentation  = .TRUE.,              ! enable cloud water sedimentation
   nc_const                                = 200.0E6,          ! cloud droplet concentration
   call_microphysics_at_all_substeps = .FALSE.,   ! microphysics every sub-timestep disabled

!
!-- steering of ice phase
!-----------------------------------------------------------------------------
   microphysics_ice_phase      = .TRUE.,          !  turn on ice microphysics
   in_init                                  = 4000.0,       ! set ice nucleii concentration
   ice_crystal_sedimentation  = .TRUE.,          ! turn on ice sedimentation
   snow                                     = .TRUE.,          ! turn on prog. quantities for snow
   graupel                                  = .TRUE.,          ! turn on prog. quantities for graupel
   start_ice_microphysics       = 0.0,              ! start with ice microphysics at model start

/ ! end of bulk cloud parameter namelist
```

Bulk cloud model

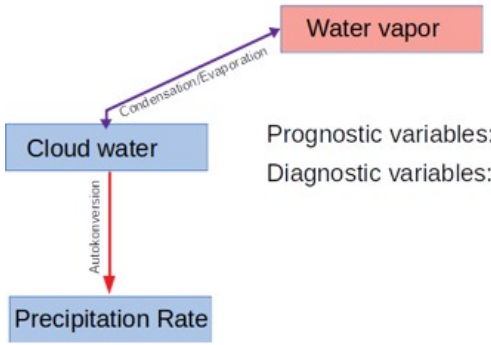
Overview of Schemes

Saturation Adjustment



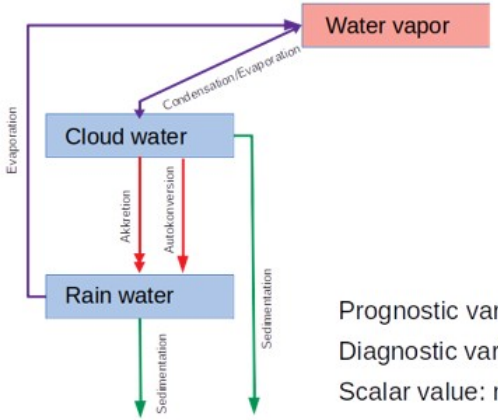
Prognostic variables: q
Diagnostic variables: qc

Kessler



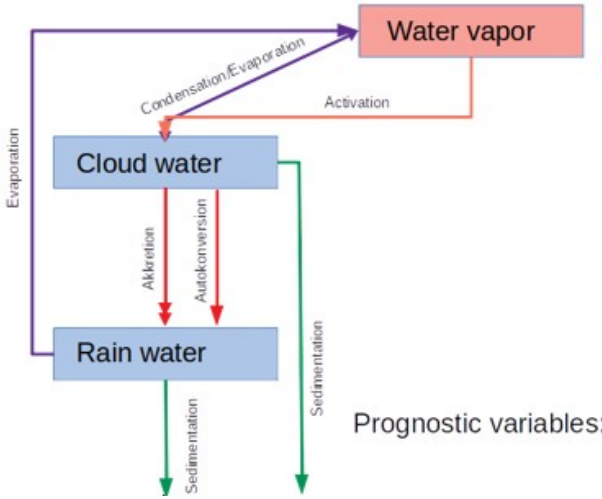
Prognostic variables: q
Diagnostic variables: qc

Seifert and Beheng



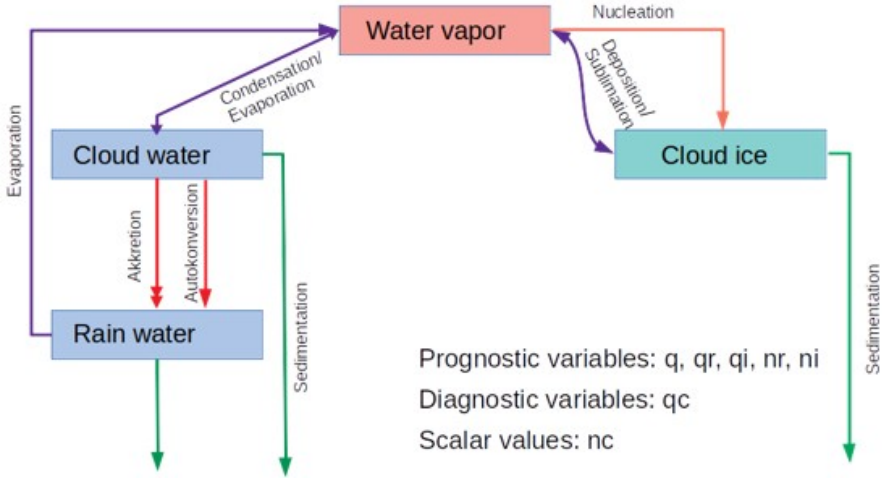
Prognostic variables: q, qr, nr
Diagnostic variables: qc
Scalar value: nc

Morrison



Prognostic variables: q, qr, nr, qc, nc

Seifert and Beheng + Microphysics ice phase

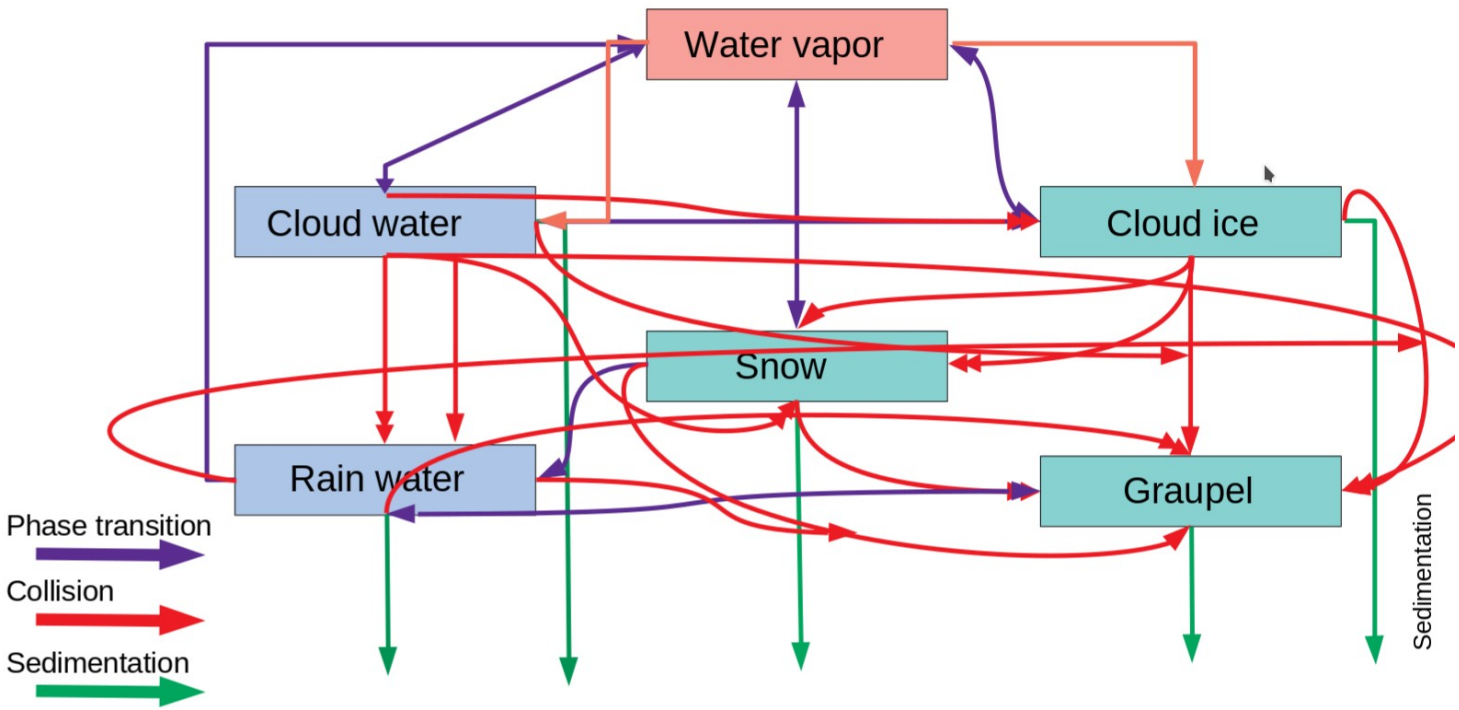


Prognostic variables: q, qr, qi, nr, ni
Diagnostic variables: qc
Scalar values: nc

Bulk cloud model

Overview of Schemes

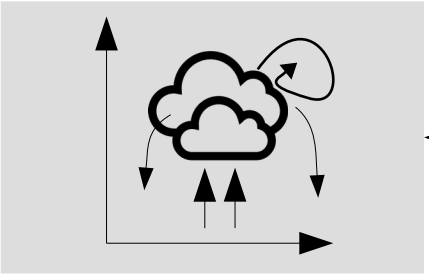
Seifert and Beheng/Morrison + microphysics_ice + snow and graupel



Prog. Quantities: qc, nc, qr, nr, qi, ni, qs, ns, qg, ng

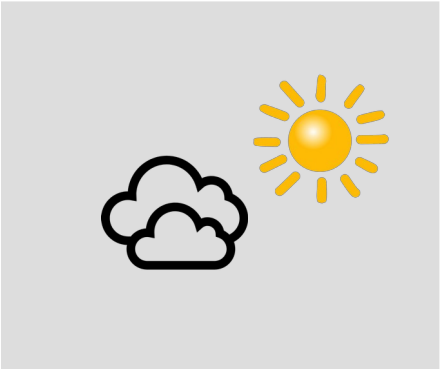
Bulk cloud model

Coupling & Limitations

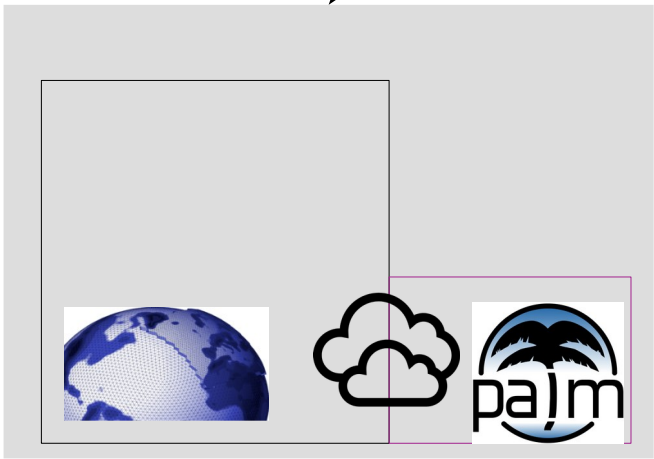


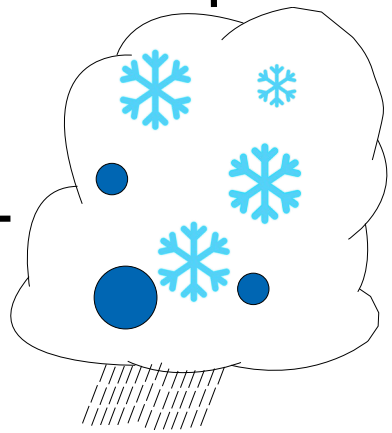
- Complete coupling with dynamics

- Coupled with radiation model (RRTMG)
- but only within one column, i.e., solely perpendicular shading is considered

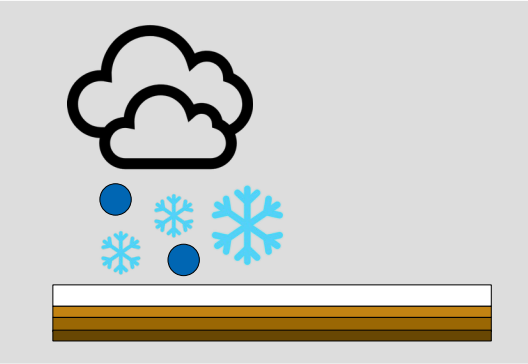


- Under development: Coupling with mesoscale models





- Precipitation is coupled to land surface model (adds on vegetation or water reservoir)
- Ice surface scheme is under development



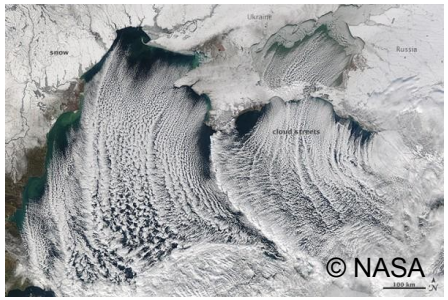
Bulk cloud model

Applications

- Simulations of fog (*e.g., Maronga and Bosveld, 2017, Schwenkel and Maronga 2019*)
- Simulations of trade wind cumuli (*e.g., Riechelmann et al., 2012*)
- Simulations of arctic mixed-phased stratocumulus clouds
- Simulations of cloud streets during cold air outbreaks (*e.g., Gryschka et al., 2014*)
- Simulation of an idealized thunderstorm

© Bob Rauber

© Patrick Kelley

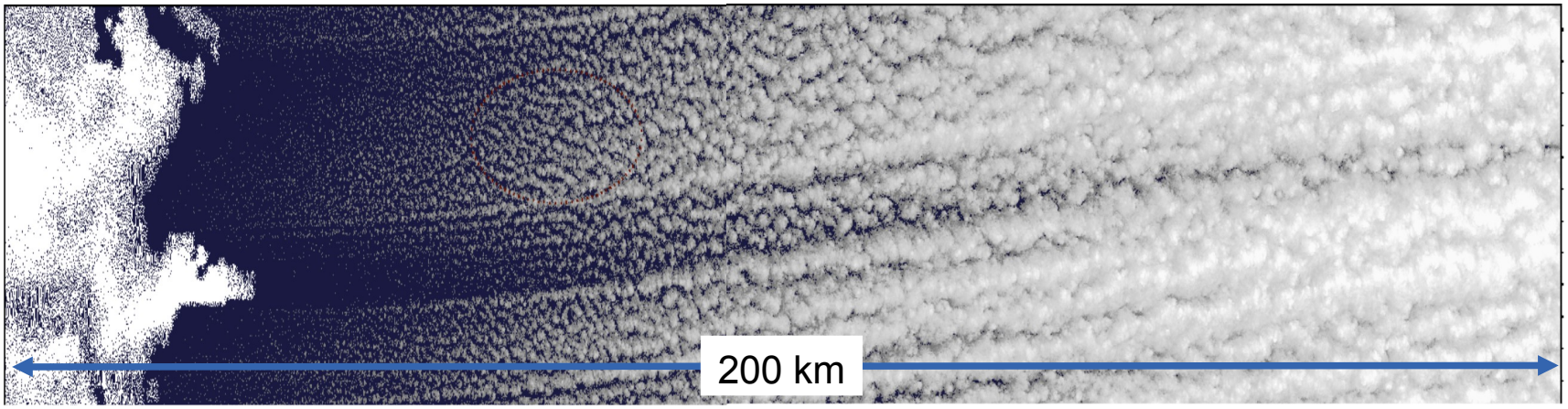

© NASA

Bulk cloud model

Applications

Thunderstorm simulation by J. Schwenkel (visualized by H. Knoop)

Simulation of Cold-Air-Outbreak and Cloud Streets by M. Gryschka



Bulk cloud model

Summary

- Computational efficient module to consider clouds with different level of detail
- Two-moment scheme allows representation of complex cloud processes (parameterized)
- A more sophisticated cloud microphysical representation can achieved with the LCM
- Coupling to other modules allows simulating more complex cases (quasi-realistic cases)

Bulk cloud model

The End